

scGPT Human Validation — Consensus & Coordinator

Purpose

Coordinate the blinded validation, preserve both locked human records, and derive a consensus human reference standard. The machine-only crosswalk is intentionally kept outside Google Docs.

Reviewer workspaces

Reviewer 1 workspace:  scGPT Human Validation — Reviewer 1 (Blinded)

Reviewer 2 workspace:  scGPT Human Validation — Reviewer 2 (Blinded, paginated)

Workflow

Assignment: share each reviewer document only with its assigned reviewer. Do not share this consensus document until both independent formal reviews are locked.

Calibration: both reviewers complete CAL-01 to CAL-03 independently, discuss them, and freeze the codebook before formal validation.

Independent review: reviewers complete all 50 HV records in their own different orders without viewing the other document.

Lock: record completion outside the document and prevent answer changes before reconciliation.

Reconciliation: copy both locked records into the matching HV row below; discuss disagreements only; enter the consensus result.

Unresolved reference: use REFERENCE_UNRESOLVED when the two reviewers cannot agree. Never force a label.

Analysis boundary: use the private local sampling crosswalk only after consensus is frozen. The 40 machine-classified studies support weighted accuracy metrics; the 10 abstention states are a separate process audit.

Locked record format

Citation validity: Yes / No / Unclear

Method executed: Yes / No / Unclear

Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear

Biological insight: Yes / No / Unclear

Evidence sufficiency: Sufficient / Incomplete / Citation not located
Supporting evidence: exact quote plus section, page, paragraph, figure, or table locator.
Consensus status: AGREED / RECONCILED / REFERENCE_UNRESOLVED

Consensus table

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
CAL-01 CALIBRATION scMGCL: accurate and efficient integration representation of single-cell multi-omics data 2025 Bioinformatics DOI: 10.1093/bioinformatics/btaf392 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
CAL-02 CALIBRATION Genome language modeling (GLM): a beginner's cheat sheet 2025 Biology Methods and Protocols DOI: 10.1093/biomethods/bpaf022 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
CAL-03 CALIBRATION Building a learnable universal coordinate system for single-cell atlas with a joint-VAE model 2024 Communications Biology DOI: 10.1038/s42003-024-06564-0 Article page Full text 1 Full text 2	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-001 FORMAL VALIDATION Generative genomics accurately predicts future experimental results 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.09.08.674753 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-002 FORMAL VALIDATION PertDiffBench: Benchmarking Diffusion Models for Single-Cell Perturbation Response Prediction 2026 DOI: 10.64898/2026.06.13.732013 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-003 FORMAL VALIDATION sciLaMA: A Single-Cell Representation Learning Framework to Leverage Prior Knowledge from Large Language Models 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.01.28.635153 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-004 FORMAL VALIDATION Discovering Candidate Anti-Aging Perturbations Using a Foundation Model for Gene Expression 2025 International Journal of Molecular Sciences DOI: 10.3390/ijms262411977 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-005 FORMAL VALIDATION A multimodal cell-free RNA language model for liquid biopsy applications 2025 Nature Machine Intelligence DOI: 10.1038/s42256-025-01148-x Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-006 FORMAL VALIDATION Artificial Intelligence agents for biological research: a survey 2026 Briefings in Bioinformatics DOI: 10.1093/bib/bbag075 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-007 FORMAL VALIDATION Benchmarking gene expression reconstruction from single-cell latent representations 2026 DOI: 10.64898/2026.06.15.731445 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-008 FORMAL VALIDATION Towards Cross-Sample Alignment for Multi-Modal Representation Learning in Spatial Transcriptomics 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.03.02.709002 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-009 FORMAL VALIDATION iPro-CSAF: identification of promoters based on convolutional spiking neural networks and spiking attention mechanism 2025 PeerJ Computer Science DOI: 10.7717/peerj-cs.2761 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-010 FORMAL VALIDATION evoCancerGPT: Generating Zero-Shot Single-Cell and Single-Sample Cancer Progression Through Transfer Learning 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.02.12.705621 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-011 FORMAL VALIDATION NanoCellAnnotator: Formalizing Expert Cell Type Annotation with Large Language Models 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.21.728965 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-012 FORMAL VALIDATION MOSAIC: Intra-tumoral heterogeneity characterization through large-scale spatial and cell-resolved multi-omics profiling 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.05.15.654189 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-013 FORMAL VALIDATION Integrating In silico perturbation with multilayer omics to decode regulatory networks in cancer immunity: a new frontier in precision oncology 2026 Frontiers in Immunology DOI: 10.3389/fimmu.2026.1857447 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-014 FORMAL VALIDATION Residual-stream geometry of single-cell foundation models carries incremental gene-regulatory signal across tissues 2026 BMC Bioinformatics DOI: 10.1186/s12859-026-06538-5 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-015 FORMAL VALIDATION scGPT-spatial: Continual Pretraining of Single-Cell Foundation Model for Spatial Transcriptomics 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.02.05.636714 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-016 FORMAL VALIDATION Investigation of cell development and tissue structure network based on natural Language processing of scRNA-seq data 2025 Journal of Translational Medicine DOI: 10.1186/s12967-025-06263-2 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-017 FORMAL VALIDATION CellMemory: hierarchical interpretation of out-of-distribution cells using bottlenecked transformer 2025 Genome biology DOI: 10.1186/s13059-025-03638-y Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-018 FORMAL VALIDATION Cellfm-datasets: A Unified Data Infrastructure for Single-Cell and Spatial Transcriptomics Foundation Model Pretraining 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.11.731508 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-019 FORMAL VALIDATION Toward automated and explainable high-throughput perturbation analysis in single cells 2025 Patterns DOI: 10.1016/j.patter.2025.101228 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-020 FORMAL VALIDATION Challenges in AI-driven Biomedical Multimodal Data Fusion and Analysis 2025 Genomics Proteomics & Bioinformatics DOI: 10.1093/gpbjnl/qzaf011 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-021 FORMAL VALIDATION scaLR: a low-resource deep neural network-based platform for single cell analysis and biomarker discovery 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf243 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-022 FORMAL VALIDATION PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.02.26.707870 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-023 FORMAL VALIDATION Genetic Convergence Analysis of CRISPR Perturbations Deciphers Gene Functional Similarity 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.11.13.688060 Article page Full text 1 Full text 2	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-024 FORMAL VALIDATION Scouter predicts transcriptional responses to genetic perturbations with large language model embeddings 2025 Nature Computational Science DOI: 10.1038/s43588-025-00912-8 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-025 FORMAL VALIDATION Hybrid Attention-Enhanced Regularized Convolutional Autoencoder for Robust EEG Feature Extraction in Seizure Prediction 2025 IEEE Access DOI: 10.1109/access.2025.3608302 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-026 FORMAL VALIDATION Cancer-associated fibroblasts enhance colorectal cancer lymphatic metastasis via CLEC11A/LGR5-mediated WNT pathway activation 2025 Journal of Clinical Investigation DOI: 10.1172/jci194243 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-027 FORMAL VALIDATION Bioinformatics Copilot 2.0 for Transcriptomic Data Analysis 2024 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.08.15.607673 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-028 FORMAL VALIDATION Single-cell sequencing in precision management of chronic myeloid leukemia 2026 Journal of Translational Genetics and Genomics DOI: 10.20517/jtgg.2026.42 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-029 FORMAL VALIDATION Elucidating the Design Space of Generative Models for Single-Cell Perturbation Prediction 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.15.732063 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-030 FORMAL VALIDATION A Foundation Model Should Not Flatten Pharmaceutical Evidence across Molecules, Sequences, Omics, Images, and Scientific Text 2024 Pharmacophore DOI: 10.51847/rbfe0o1wzp Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-031 FORMAL VALIDATION C5aR1 inhibition reprograms tumor associated macrophages and reverses PARP inhibitor resistance in breast cancer 2024 Nature Communications DOI: 10.1038/s41467-024-48637-y Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-032 FORMAL VALIDATION OCOO-T : A SIMPLE AND SCALABLE VIRTUAL CELL MODEL FOR TRANSCRIPTIONAL PERTURBATION RESPONSE PREDICTION 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.08.731000 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-033 FORMAL VALIDATION scCCVGBen for benchmarking of single-cell representation learning anchored on a centroid-coupled variational graph attention autoencoder across scRNA-seq and scATAC-seq 2026 Frontiers in Genetics DOI: 10.3389/fgene.2026.1822168 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-034 FORMAL VALIDATION GET: a foundation model of transcription across human cell types 2023 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2023.09.24.559168 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-035 FORMAL VALIDATION Revealing the Best Strategies for Rare Cell Type Detection in Multi-Sample Single-Cell Datasets 2025 Genes DOI: 10.3390/genes17010031 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-036 FORMAL VALIDATION CART-GPT: A T Cell-Informed AI Linguistic Framework for Interpreting Neurotoxicity and Therapeutic Outcomes in CAR-T Therapy 2025 DOI: 10.1101/2025.08.08.669387 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-037 FORMAL VALIDATION Tahoe-x1: Scaling Perturbation-Trained Single-Cell Foundation Models to 3 Billion Parameters 2025 DOI: 10.1101/2025.10.23.683759 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-038 FORMAL VALIDATION cspray: Distributed Single Cell Transcriptome Analysis 2026 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.02.06.704110 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-039 FORMAL VALIDATION scGNN+: Adapting ChatGPT for Seamless Tutorial and Code Optimization 2024 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.09.30.615735 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-040 FORMAL VALIDATION Clifti-GPT: Privacy-preserving federated fine-tuning and transferable inference of foundation models on clinical single-cell data 2025 DOI: 10.21203/rs.3.rs-7917089/v1 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-041 FORMAL VALIDATION Nicheformer: a foundation model for single-cell and spatial omics 2025 Nature Methods DOI: 10.1038/s41592-025-02814-z Article page Full text 1 Full text 2	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-042 FORMAL VALIDATION Foundation model reveals the shared organization of transcription and topologically associating domains 2026 Cell Systems DOI: 10.1016/j.cels.2026.101675 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-043 FORMAL VALIDATION DECIPHER for learning disentangled cellular embeddings in large-scale heterogeneous spatial omics data 2025 Nature Communications DOI: 10.1038/s41467-025-63140-8 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-044 FORMAL VALIDATION CellAgent: LLM-Driven Multi-Agent Framework for Natural Language-Based Single-Cell Analysis 2024 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.05.13.593861 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-045 FORMAL VALIDATION Opportunities for artificial intelligence and synthetic biology in designing living drug delivery systems 2026 Advanced Drug Delivery Reviews DOI: 10.1016/j.addr.2026.115883 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-046 FORMAL VALIDATION AUPRC: a metric for evaluating the performance of in-silico perturbation methods in identifying differentially expressed genes 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf426 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-047 FORMAL VALIDATION A technical review of multi-omics data integration methods: from classical statistical to deep generative approaches 2025 Briefings in Bioinformatics DOI: 10.1093/bib/bbaf355 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:

Article and approved evidence links	Reviewer 1 locked record	Reviewer 2 locked record	Consensus human reference
HV-048 FORMAL VALIDATION scBaseCount: an AI agent-curated, uniformly processed, and autonomously updated single cell data repository 2025 bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.02.27.640494 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-049 FORMAL VALIDATION Routine FFPE sections support clinically compatible single-nucleus transcriptomics across six human cancer types 2026 DOI: 10.64898/2026.07.23.740343 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note:
HV-050 FORMAL VALIDATION Context-dependent utility and robustness of pretrained single-cell foundation model representations across analytical tasks 2026 DOI: 10.64898/2026.06.18.733285 Article page Full text 1	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Notes:	Consensus status: Citation validity: Method executed: Primary use: Biological insight: Evidence sufficiency: Supporting quote + locator: Resolution note: